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Gravimetric study of the shallow basaltic plumbing system of Tenerife, Canary Islands

S. Sainz-Maza Aparicio[a,c], J. Martí[b,*], F.G. Montesinos[c,d], A. Borreguero Gómez[a], J. Pereda de Pablo[e], P. Vaquero Fernández[f], M. Calvo García-Maroto[a]

[a] Observatorio Geofísico Central (IGN), C/ Alfonso XII, 3, 28014 Madrid, Spain
[b] Institute of Earth Sciences Jaume Almera, ICTJA-CSIC, Lluís Sole i Sabaris s/n, Barcelona, Spain
[c] Grupo Geodesia, UCM, Madrid, Spain
[d] Facultad de Matemáticas, Universidad Complutense de Madrid, Madrid, Spain
[e] Centro Geofísico de Canarias, Instituto Geográfico Nacional, C/La Marina, 20, 38001 Santa Cruz de Tenerife, Spain
[f] Instituto Geográfico Nacional, C/del Gral. Ibáñez de Íbero, 3, 28003 Madrid, Spain

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ABSTRACT

Recent studies on intraplate volcanic islands reveal that magma ascends sub-vertically through these volcanic edifices, following well-defined zones in the lithospheric mantle and oceanic crust, and occasionally suffering lateral propagation along rift zones when it reaches very shallow depths. Visualizing the internal structure of the shallowest part of such magma plumbing systems is crucial to understanding magma transport and the location of basaltic volcanism in these volcanic environments. Tenerife (Canary Islands) is a very good example of an intraplate volcanic island, in which the distribution of basaltic volcanism includes well-defined rift systems, and wider monogenetic fields where cinder cones form clusters along different tectonic alignments. In order to characterize the structure of the uppermost part of the basaltic plumbing system of Tenerife, we conducted a detailed gravimetric study of the north-western, central, and southern sectors of the island. We developed a 3D density contrast model of the subsoil based on the application of a genetic algorithm for the inversion of gravity data. The results of our inverse modelling suggest that the observed gravimetric anomalies below the north-western rift zone are shallow and aligned parallel to this structure. This is consistent with a shallow character for the rift system and a lateral migration of magma along it that comes from a deeper zone located towards the east of this rift zone. Conversely, the anomalies located in the south are also shallow but are distributed along more than one alignment and are connected to much deeper sub-vertical similar density anomalies. These results support the existence of several main channels at the north and south of the island that allow the ascent of deep basaltic magma to shallower zones of Tenerife. There, magma may be either caught by the rifts systems, being distributed laterally and occasionally erupting, or may continue its ascent to the surface using a secondary system of shallow fractures that dominates the southern part of Tenerife. The results of our inverse model also offer new insights on others aspects of the internal structure of Tenerife, such as the distribution of the basaltic shield or that of Las Cañadas caldera.

1. Introduction

Intraplate volcanic islands are almost exclusively associated with alkaline basaltic volcanism (e.g., Niu et al., 2011; Kroppers, 2011; Deschamps et al., 2011), while cases in which differentiated magmas are also common in these volcanic environments are rare (e.g., Tenerife, Martí et al., 2008). The distribution of basaltic volcanism on ocean volcanic islands is characterized by eruptive fissures or parallel rows of elongated cinder cones and dyke swarms. These are interpreted as rifts systems, in comparison with the proper extensional structures of intraplate continental volcanism (Walker, 1999). However, the presence of monogenetic basaltic fields where volcanic centers show a much wider distribution and seem to follow an apparent random distribution or are grouped into forming different clusters, has also been described in such volcanic environments (e.g., Lockwood and Lipman, 1987; Villeneuve and Bachèlery, 2006; Dóniz et al., 2008; Kereszturi et al., 2013; Michon et al., 2015). Therefore, different structural controls may determine the distribution of basaltic volcanism on ocean

* Corresponding author.
E-mail address: joan.marti@ictja.csic.es (J. Martí).

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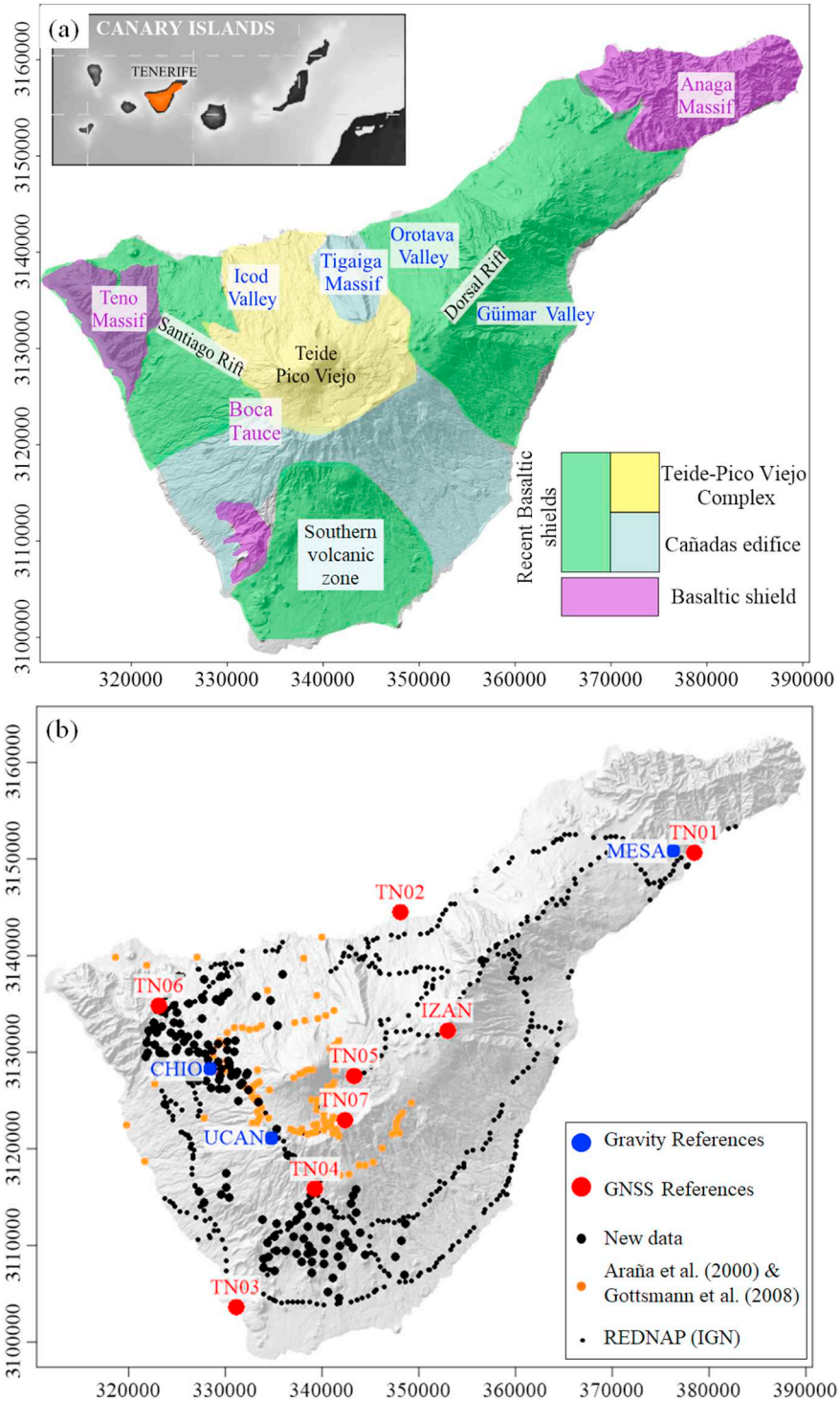

Fig.1. (a) Main geological features of Tenerife (Canary Islands). (b) Distribution of the GPS and gravity observations used in this work.

volcanic islands, despite the fact that the exact structural controls and internal architecture of these basaltic plumbing systems are still not well constrained.

Rift systems are assumed to be associated with extensional stresses and are interpreted as being controlled by deep crustal structures or by tectonic activity such as faulting and plate motion, or as the product of the inflation of the volcano due to mantle upwelling or volcanic spreading (Walker, 1999; Carracedo, 1994; Walter and Troll, 2003; Walter, 2003; Chaput et al., 2014; Michon et al., 2015). Volcanic rift areas are usually described as having originated by persistent dyke swarm injections with particular orientations that penetrate the weakest zones of the upper crust under a local extensional stress regime (Walker, 1999), although other configurations are possible and will depend on topography, edifice shape, and regional tectonic control (Acocella and Neri, 2009). At the surface, these structures appear as ridges occupied by rows of aligned cones and fissures. Particular cases of this type of volcanism include the construction and growth of several oceanic volcanic islands such as Hawaii, Reunion Islands, and the Canary Islands (Fiske and Jackson, 1972; Carracedo, 1994; Michon et al., 2015). Conversely, in monogenetic basaltic fields, volcanic

centers tend to form clusters of different ages, arranged following different alignments depending on temporal changes in regional and local tectonic controls (e.g., Cassidy et al., 1999; Cassidy and Locke, 2010; Le Corvec et al., 2013). Monogenetic volcanic fields are common in continental environments (see Németh, 2010), but similar arrangements of monogenetic volcanic vents are also observed on intraplate volcanic islands such as Hawaii (Lockwood and Lipman, 1987), La Reunion (Villeneuve and Bachèlery, 2006; Michon et al., 2015) or Tenerife (Dóniz et al., 2008; Kereszturi et al., 2013).

The Canary Islands are a particular case of intraplate volcanism (Araña and Ortiz, 1991), where both basaltic rift systems and wider monogenetic fields are present (Troll and Carracedo, 2016). Processes linked to rift development have played an important role in driving basaltic magma to the surface but also in shaping the morphology of these islands, both in terms of their vertical development and the results of the giant landslides that are the most striking surface features of the youngest islands (Carracedo, 1994, 1996; Gee et al., 2001; Walter and Troll, 2003; Walter et al., 2005; Manconi et al., 2009). On most islands, basaltic volcanism also occurs outside the rift zones, but has received less attention.

An excellent example of how basaltic rifts systems and wider monogenetic fields coexist on an intraplate volcanic island is Tenerife (Canary Islands). In addition, basaltic volcanism shares the island with the stratovolcano Teide-Pico Viejo. This combination has given rise to a complex interaction between basaltic and phonolitic magmatic systems, depicted in a large variety of eruptive styles (Martí et al., 2008) that are not found on other intraplate volcanic islands. This complex interaction between deep and shallow magmatic systems makes it difficult to understand the dynamics of Tenerife volcanism and, in particular, the structure of its plumbing system. This is the reason why some of the most significant aspects of the Tenerife basaltic volcanism, such as the number of rift systems and whether they are deep or shallow structures, are still under debate (see Carracedo, 1994, 1996; Martí et al., 1996; Walter, 2003; Geyer and Martí, 2010).

In order to help resolve such important questions we used 3D gravity-inversion procedures to image the internal structure of the shallowest part of the basaltic plumbing system of Tenerife and to explain how it controls magma transport and the location of basaltic volcanism. To accomplish such an objective we compared the internal density structures of two zones that host recent basaltic volcanism: i) the Santiago del Teide rift area (NW rift), in which most of the cones are aligned parallel to the same structure (Carracedo et al., 2007), and ii) southern Tenerife, where there is an older monogenetic volcanic field, in which volcanic cones form several clusters not following a unique alignment (Bryan et al., 1998; Kröchert and Buchner, 2008; Buchner and Kröchert, 2009; Geyer and Martí, 2010). Our results allow highlighting the main differences in terms of location and depth of the gravity anomalies that can be attributed to the basaltic plumbing systems of these two monogenetic zones, and also shed new light on some other relevant aspects of the Tenerife interior.

2. Geological setting

2.1. General features

The Canary Islands Archipelago is located in the Atlantic Ocean, near the southern coast of Morocco (African plate). It consists of a group of seven major islands and six islets, the highest (3718 m) and largest (2034 km^2) of which is Tenerife (Fig. 1a). The age sequence and spatial distribution of these islands and nearby seamounts is more complex than other intraplate island groups such as the Hawaiian chain, and so a variety of models—for instance, a dominant lithospheric control of magmatism (e.g., Anguita and Hernan, 1975; Araña and Ortiz, 1991), a mantle plume origin (e.g., Morgan, 1971; Hoernle et al., 1991; Hoernle and Schmincke, 1993; Carracedo et al., 1998), or a combination of both (Anguita and Hernán, 2000)—have been proposed to describe Canarian magmatism.

The geological evolution of Tenerife involves the construction of two main volcanic complexes (Martí et al., 2008), namely, a basaltic shield complex (from 12 Ma–present; Abdel-Monem et al., 1972; Ancochea et al., 1990; Thirlwall et al., 2000) and a central complex (from 4 Ma–present; Fúster et al., 1968; Araña, 1971; Ancochea et al., 1990; Martí et al., 1994). The basaltic shield complex is mostly submerged and makes up about 90% of the volume of the island, whose subaerial construction continues at present through the two rift zones (Santiago and Dorsal Rift Zones).

The central complex (Fig. 1a) consists of the Cañadas edifice (from 4 to 0.18 Ma), a composite volcano characterized by abundant explosive eruptions of highly evolved phonolitic magmas, and the active twin Teide–Pico Viejo stratovolcanoes (0.18 Ma–present). These stratovolcanoes have evolved from basaltic to phonolitic and their activity has mostly been effusive. The caldera of Las Cañadas (Fig. 1a), which hosts the Teide–Pico Viejo stratovolcanoes, truncated the Cañadas edifice and has been transformed by several episodes of vertical collapse, occasionally associated with the lateral collapse of the volcano's flanks (Martí et al., 1994, 1997; Martí and Gudmundsson, 2000).

Several geophysical studies using techniques such as gravimetry (Ablay and Kearey, 2000; Araña et al., 2000; Gottsmann et al., 2008) and geomagnetism (Araña et al., 2000; Blanco-Montenegro et al., 2011), distinct seismic methodologies (Canales et al., 2000; García-Yeguas et al., 2012; Prudencio et al., 2013), resistivity methods (Pous et al., 2002; Coppo et al., 2008), magnetotelluric exploration (Piña-Varas et al., 2014) and self-potential techniques (Villasante-Marcos et al., 2014) have provided relevant information on the internal structure of Tenerife. The exploration of the 1500 km of water galleries drilled into the flanks of the island has facilitated establishing the internal stratigraphy of certain parts of the island (Coello, 1973; Boulesteix et al., 2012; Iribarren, 2014). All this information highlights the fact that Tenerife is essentially a basaltic shield structure on which a central phonolitic complex has developed. This basaltic shield is still visible at the three corners of the island (Teno, Anaga, and Roque del Conde) (Fig. 1), while the nature of the central part, now mostly covered by Las Cañadas caldera and the Teide-Pico Viejo products, is revealed by a high-density core extending in depth below the products of Las Cañadas (Ablay and Kearey, 2000; Araña et al., 2000; Gottsmann et al., 2008). The uppermost part of this dense body is an old buried volcano named Boca Tauce (Ablay and Kearey, 2000) (Fig. 1a). Additional high-density structures have been also inferred under the Teno, Anaga and Roque del Conde massifs (Fig. 1a). Las Cañadas has a negative gravity anomaly (Ablay and Kearey, 2000; Araña et al., 2000; Gottsmann et al., 2008) and its three main depressions, corresponding to the three overlapping caldera episodes described by Martí et al. (1994), have been clearly defined by magnetotelluric and electrical methods (Pous et al., 2002; Coppo et al., 2008; Villasante et al., 2014).

Finally, available geophysical data, combined with stratigraphic data from the water galleries and the borehole data can be used to infer that the caldera fill corresponds to deposits from Las Cañadas and to at least 540 m of lavas and pyroclasts from Teide-Pico Viejo (Iribarren, 2014). The total depth at which the caldera floor should be located is known. However, a minimum estimate based on borehole and geophysical data, and on exposed parts of the caldera wall suggests that up to 1200 m of the caldera collapsed (Iribarren, 2014). In a similar way, the three main landslide valleys, Güimar, La Orotava, and Icod (Fig. 1a), have all been in-filled with products from Las Cañadas, the rifts and Teide-Pico Viejo, the exception being the Icod valley, which does not have any Las Cañadas products in its interior (Coello, 1973; Iribarren, 2014).

2.2. Surface distribution of recent basaltic volcanism

The structural and morphological features visible on the surface of the Santiago del Teide and Dorsal Rift Zones (Fig. 1a) are very similar.

In both cases, the alignments of the basaltic cones and dikes follow their characteristic main structural directions, that is, a NW direction in the case of the Santiago del Teide rift and a NE direction in the case of the Dorsal rift. Volcanic activity along these two rifts seems to have been constantly present during the formation of the island, as suggested by the age of the oldest rocks as well as the coincidence in the orientation of the dykes in the rifts and the remains of the basaltic shield. This also suggests that the rift zones inherited their orientation from the main regional tectonic trends present in the basement of the Canary Islands (Geyer and Martí, 2010). Intrusive activity along these rift zones could have contributed to some of the gravitational landslides that affected the flanks of Tenerife during its construction (Carracedo, 1994; Carracedo et al., 2010). In addition, a number of paleomagnetism studies have investigated the flow emplacement of dykes in the rift zones and in the adjacent oldest part of the island, and provide evidence for the lateral flow of magma in all cases (Soriano et al., 2008; Delcamp et al., 2010, 2014).

Moreover, a monogenetic basaltic volcanic field developed on the southern flank of the island that was mostly active during the final episodes of the construction of the edifice of Las Cañadas (Bryan et al., 1998; Dóniz et al., 2008; Kereszturi et al., 2013). This zone has also been interpreted as a N-S-oriented rift zone by Carracedo (1994). However, there are significant structural differences between the southern monogenetic field and with the Santiago del Teide and Dorsal rifts. For example, apart from the differences in the age of the volcanic cones clustering (Kröchert and Buchner, 2008; Buchner and Kröchert, 2009), in the south there are two preferential alignments (NNW and NE), along which most of the cones are located (Bryan et al., 1998; Dóniz et al., 2008; Kröchert and Buchner, 2008; Buchner and Kröchert, 2009; Kereszturi et al., 2013). Geyer and Martí (2010) suggested that the monogenetic basaltic field in southern Tenerife is related to the extensional stress field provoked by the combined effects of the two rifts and is not a third true rift zone like the Santiago del Teide and Dorsal rifts. This coincides with recent geophysical studies (Blanco-Montenegro et al., 2011; García-Yeguas et al., 2012) that also cast doubt on the existence of a southern rift; these authors found the same alignment at depth as those inferred from the surface by Buchner and Kröchert (2009) and Geyer and Martí (2010), but found no evidence of the NS rift postulated by Carracedo (1994).

3. Data acquisition

In order to develop a new underground density model for the northwestern and southern sectors of Tenerife, 159 new relative gravity and GPS observations were taken in 2010–2013 (see Supplementary material), which were then employed in combination with 418 previous readings from two other sources: gravity data obtained during the high-precision levelling network development, which was conducted by the Spanish National Geographic Institute (IGN) and data from Araña et al. (2000) and Gottsmann et al. (2008). Thus, the coverage of both sectors was improved and we were thus able to generate a more detailed map of the residual anomalies (Fig. 2). We employed a 3D gravity-inversion procedure based on a genetic algorithm (Montesinos et al., 2005; Montesinos et al., 2006) to create a detailed density contrast model of structures at depth.

The most recent relative gravity data were obtained using a Lacoste & Romberg LR-G301 zero-length spring gravimeter (© Lacoste, 1988), which provides an accuracy of over 15 μGal (1 μGal = 10^{-8} m/s^2) and operates with very low drift (around 20 μGal/day). Surveys were divided into daily itineraries consisting in closed loops that covered at least 7 different points per day around a base station. At any itinerary, some points were revisited at least twice in order to control the potential tares and the drift of the gravimeter. Over a period of 15 min, 3–6 gravity measurements were taken at each point. The base station was located at the geodetic benchmark CHIO (28.2689°N, 16.7495°W) at 1591 m a.s.l (see Fig. 1b). In addition, an absolute gravity value was obtained at the old Las Mesas Geomagnetic Observatory (MESA) (Fig. 1b) with an FG5#211 gravimeter (http://www.microglacoste.com/fg5.php). The relative gravity value between MESA and CHIO was measured several times with the LR-G301 to obtain the absolute gravity value at CHIO. This enabled us to acquire the absolute gravity values for all the stations used in this study.

The new gravity measurements were made simultaneously with the GNSS observations. We used two models of dual-frequency GPS receivers (Trimble 4000SSi in 2010 and 2011, and Trimble R4 in 2013). Every measurement lasted for 10 or 15 min, with a sampling frequency of 1 Hz. New GNSS observations were post-processed jointly with the permanent geodetic Spanish IGN network on Tenerife (see Fig. 1(b)).

4. Data processing

GPS and field gravity data were processed to compute the Bouguer anomalies. The GNSS coordinates of the new observation points were obtained using the Trimble Total Control® software (http://www.trimble.com/). The ellipsoidal to orthometric height transformations were made using the geoid undulation obtained from the model EGM2008-REDNAP (an adaptation of the global geoid model EGM2008 of the National Geospatial Intelligence Agency to the vertical reference system in Spain, REDNAP).

The first step in the gravity data processing was to reduce the new gravity observations relative to the CHIO station to absolute gravity values at ground level by applying standard corrections (Torge, 1989) (see Eq. (1)). Thus, relative gravity values were transferred from instrumental to ground height using the theoretical vertical gravity gradient of −308.6 μGal/m. An atmospheric correction based on the US Standard Atmosphere model (NOAA/NASA/USAF, 1976) was also applied. This correction compensates gravity attraction of the air mass differences for each station, which depends mostly on their respective heights. The gravimeter mechanical drift was adjusted using simple linear fits. Earth tides and ocean loading were modelled and removed from data in accordance with the Wenzel catalogue and the FES04 Global model, respectively (Hartmann and Wenzel, 1995; Lyard et al., 2006). Finally, the gravity difference between CHIO and MESA (where an absolute gravity value was obtained) was calculated and added to all measurements to provide the absolute gravity value at each station (see Eq. (1)):

$$g_i(mGal) = g_i^0 + \Delta g_i^r - 0.3086 s_i - \Delta g_d(t_i) - \Delta g_{et}(\lambda_i, \varphi_i, h_i, t_i) - \Delta g_{ol}(\lambda_i, \varphi_i, h_i, t_i) - \Delta g_a(h_i) \quad (1)$$

where g_i^0represents the absolute gravity value at CHIO, and Δg_i^r is the relative measurement made for each i-station in relation to CHIO. The theoretical vertical gravity gradient (−308.6 μGal/m) was used to transfer the gravity values to ground height, with s_i being the gravity sensor height (m) of the i-station. The term $\Delta g_d(t_i)$ corrects the gravimeter mechanical drift and Δg_{et} and Δg_{ol} are the tidal and ocean loading corrections, respectively. The atmospheric correction (Δg_a) was computed using Eq. (2):

$$\Delta g_a(mGal) = 0.87 e^{-0.116 h_i^{1.047}} \quad (2)$$

where h_i is the orthometric height of the i-station. The second step in this procedure was to combine the new gravity data (2010 − 2013) with the corresponding data from previous data sources (IGN REDNAP; Araña et al. (2000); Gottsmann et al. (2008)) (Fig. 1b). In order to merge these data, we employed a common station used for all data sources that corresponds to the Ucanca geodetic vertex (UCAN, 28.2055°N, 16.6834°W) at 2102 m a.s.l. (see Fig. 1b). Using more than ten relative gravity connections between CHIO and UCAN, we were able to obtain an absolute gravity value for UCAN (978,983.007 ± 0.015 mGal). With this value, we then obtained the

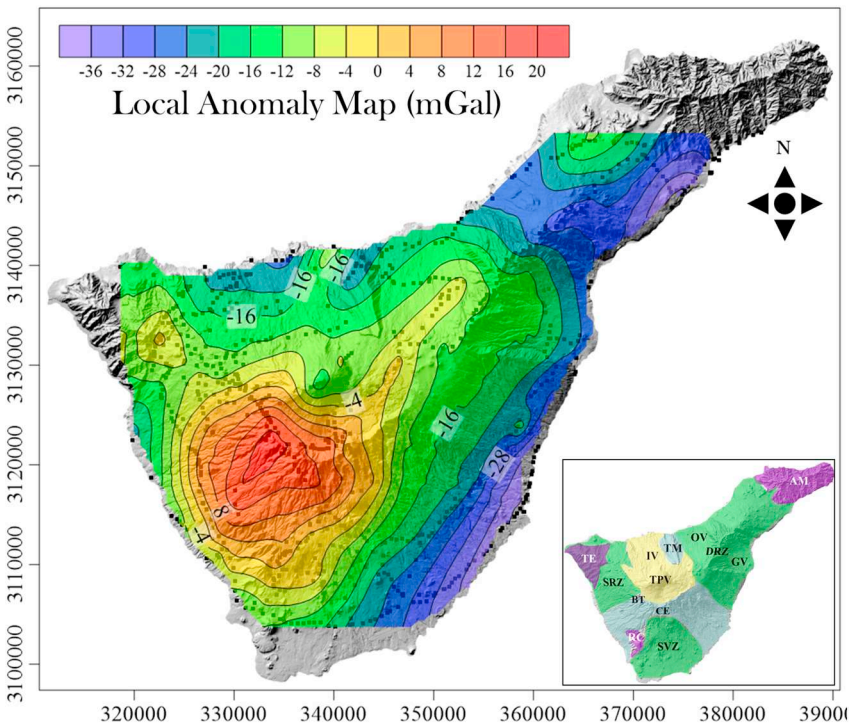

Fig. 2. Local anomaly map for onshore Tenerife, after removing a regional component from the Bouguer anomaly map. The maximum differences in the gravity anomaly are around 60 mGal. The almost concentric distribution of the anomalies around the highest values is clearly visible; its symmetry is only broken towards the NW and NE rift areas. The lower right-hand inset shows the main geological features of the island annotated by the following abbreviations: Teno Massif (TE), Anaga Massif (AM), Santiago Rift zone (SRZ), Dorsal Rift zone (DRZ), Southern volcanic zone (SVZ), Icod Valley (IV), Orotava Valley (OV), Güimar Valley (GV), Teide-Pico Viejo Complex (TPV), Las Cañadas Edifice (CE), Roque del Conde (RC).

absolute gravity values for all points of all data sources using Eq. (3).

$$g_i = g_{abs}^{UCAN} + \Delta g_i^{UCAN} \tag{3}$$

where g_i is the final absolute value in the i-station, g_{abs} is the absolute gravity value at UCAN, and Δg_i is the difference in gravity between the i-station and UCAN.

At this stage it was possible to start the computation of the Bouguer anomaly map and the corresponding residual anomaly map. The normal gravity field was modelled and subtracted from data using Somigliana's (1929) formula with the WGS84 parameters. Then, we applied the free-air correction and Bouguer plate reduction based on the GRS80 (Heiskanen and Moritz, 1985). As well, we developed a precise terrain correction for modelling topography by using prismatic bodies, and combined the terrain pattern suggested by Kane (1962) and the Nagy formulation for the prism gravitational attraction (Nagy, 1966) as in Eq. (4):

$$\Delta g_p^k = G\rho \left| \left| x_i \, ln(y_i + r_i) + y_i \, ln(x_i + r_i) - z_i \, arcsin \frac{z_i^2 + y_i^2 + x_i y_i}{(y_i + r_i)\sqrt{z_i^2 + y_i^2}} \right|_{x_1}^{x_2} \right|_{y_1}^{y_2} \Big|_{z_1}^{z_2} \tag{4}$$

where $\Delta g_p{}^k$ is the gravitational attraction of the k-prism above the p-gravity station, G is the gravitational constant, ρ is the chosen terrain density, and x_i, y_i, and z_i, (i = 1, 2) are the limiting coordinates of the corresponding k-prism on each coordinate axis in relation to the i-station. The variable r_i represents the radial distance between the k-prism center and the p-station.

The topography was modelled in three areas. The nearest area to the observed point was composed of cells with a 5 × 5 m^2 base that covered the initial 4 km^2. An intermediate area consisting of 50 × 50 m^2 cells modelled the terrain in an area of 60 × 60 km^2. Finally, for the furthest distances within a radius of 167 km, 500 × 500 m^2 cells were used. Onshore Digital Terrain Models were downloaded from the IGN Website (see http://www.ign.es/ign/layoutIn/modeloDigitalTerreno.do). Offshore areas were modelled using the bathymetry available on the GEBCO website (General Bathymetric Chart of the Oceans) (Weatherall et al., 2015). Thus, for marine areas, the terrain density was replaced by water density from the sea surface to seafloor, since in the case of islands surrounded by a deep seafloor, the use of terrain density instead of water density has been observed to disrupt the final anomaly (Nowell, 1999; Tong and Guo, 2007).

One of the most interesting features of Tenerife's volcanism is its great variety of eruptive styles and its corresponding emitted materials. This results in a great variability of volcanic product densities. According to Ablay and Kearey (2000), the density of the basaltic rocks sampled in Tenerife rift zones ranges from 1760 kg/m^3 to 2950 kg/m^3. Because of this, and given that the aim of this study was to characterize the internal structure of north-western and southern Tenerife, we selected a more appropriate mean terrain density reference based on Nettleton's (1939) methodology. This method establishes that the density value that best fits the observations minimizes the correlation between the total Bouguer anomaly and the height. We obtained a best-fit density value of 2500 kg/m^3. This contrasts with the value used by Gottsmann et al. (2008) (2200 kg/m^3), who obtained it from the averaged density of several rock samples collected at the surface. However, it is similar to the value used in other studies, such as Araña et al. (2000) (2400 kg/m^3). The main goal of the present study is to obtain detailed information of the subsurface below the most recent basaltic zones. Therefore, we only used the gravity data corresponding to these areas in order to apply the Nettleton methodology, and the results obtained refer to the mean density value in these areas.

Two further additional corrections were applied:

The regional gravimetric anomaly provides information about the structure of the deeper crust, which is not the focus of this study. In order to remove the effect generated by these regional structures, the most used methodology consists of representing the regional effect using a smoothed function, which adapts adequately to the Bouguer anomalies observed at the regional level. According to this criterion, the regional gravity field was removed from the data using a smoothed linear trend of 0.27 ± 0.03 mGal/km with an azimuth of N113°E, as indicated in Gottsmann et al., (2008), obtaining the distribution of the local gravity anomalies.

The Earth curvature effect was removed. It is particularly important at distances over 100 km, since at values higher than this, the curvature reduces the height by nearly 784 m. Thus, we used the approximation

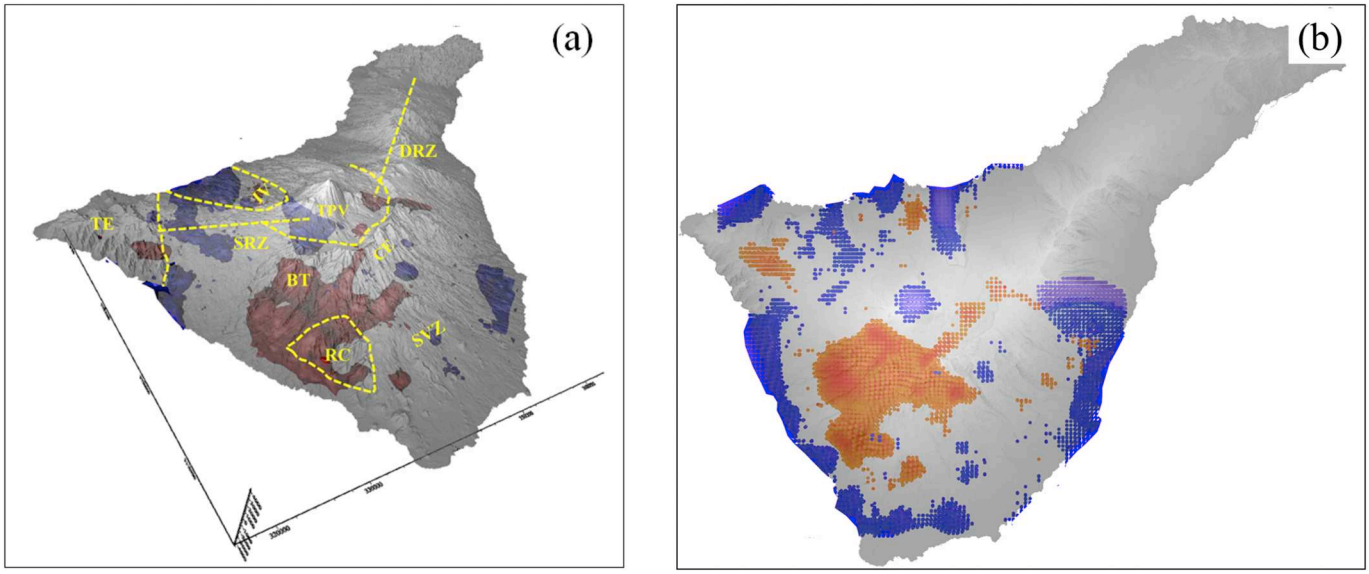

Fig. 3. Representation of the results from the gravity inversion. (a) Simplified 3D model, under Tenerife transparent relief, displaying the distribution of the main high density central body in red (density contrast > 150 kg/m^3) and the lower density contrast in blue (density contrast < 150 kg/m^3). The main morpho-structural features described in the text and in Fig. 2 are highlighted in yellow. (b) Representation of the centers of the prisms characterized by density contrasts > 150 kg/m^3. Red points represent positive density contrasts and blue points represent negative ones. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

proposed by Cogbill (1979) for the named Bullard B correction:

$$B = 14641410^{-3}h_i + 35330510^{-7}h_i^2 + 10027110^{-13}h_i^3 + 30024110^{-18}h_i^4 \quad (5)$$

where B is the Bullard correction in mGal and h_i the height of the i-station.

The application of all these procedures provided a final estimated mean precision for the local anomalies of around 200 μGal and 3–10 cm for the GPS coordinates. The final local anomaly map is shown in Fig. 2.

5. 3D gravity inversion

The 3D gravity inversion was performed using a genetic algorithm (GA) according to the procedure described in Montesinos et al. (2005). Using the gravity data from Tenerife we obtained a 3D density model of the location and geometry of the anomalous bodies under the island (Figs. 3 to 6).

The GA methodology starts with an initial prismatic partition of the subsurface volume in which each cell is described as a density contrast. Then, the algorithm generates distinct random models and performs successive modifications in an iterative process using three operators (selection, mutation and crossover). The objective function is to minimize the difference between the gravity field generated by the model and the observed gravity data, using the following equation:

$$F(\boldsymbol{m_k}) = (\boldsymbol{A}\boldsymbol{m_k} - \boldsymbol{g_{obs}} - G_m\boldsymbol{u})^T\boldsymbol{E_{ss}}^{-1}(\boldsymbol{A}\boldsymbol{m_k} - \boldsymbol{g_{obs}} - G_m\boldsymbol{u}) + \beta\boldsymbol{m_k}^T\boldsymbol{C_M}\boldsymbol{m_k} \quad (6)$$

where $\boldsymbol{g_{obs}}$ is the vector of the observed gravity anomaly in N stations, $\boldsymbol{A} = (\boldsymbol{A_{ij}})$ is an $N \times M$ matrix which defines the gravity attraction at the ith station due to the influence of the jth prism of the model, assuming as unity its density contrast. G_m represents the regional tendency defined by one constant term and $\boldsymbol{u}$ is a unitary vector. $\boldsymbol{E_{ss}}$ is a data error matrix (e.g., obtained from covariance analysis), and β is an empirical parameter corresponding to the Tikhonov regularization technique (Schwarz, 1979). Finally, the uncertainty of the model is given by the MxM·matrix $\mathbf{C_M}$. In this matrix, $\mathbf{C_M}$ defines the supposed determinability of the model parameters. This is a diagonal normalizing matrix of non-null elements, the same as the diagonal elements of $\mathbf{A^T}\boldsymbol{E_{ss}}^{-1}\mathbf{A}$, including this second term in the error function; we consider the influence of the location of the prism in the solution.

Several tests were developed aiming to select the values for the distinct algorithm input parameters. Thus, the first stage previous to applying the GA, is setting some operational parameters, describing the physical and mathematical features for the model, which provide realistic solutions at the time they contribute to the convergence of the algorithm.

Prisms size: to build our model we started with a division of the subsoil around the study area. The optimal model we found consisted of 12,182 rectangular prisms of different sizes (Fig. 3b). To determine the size of each prism, the maximum and minimum observation heights of the gravity data were taken into account, in such a way that no prism exceeds the topographic surface. In addition, the distance between the adjacent observation points was taken into account, so that each prism is represented by at least one observation of gravity at the surface. Finally, the average diameter of the network, which is approximately 35,000 m, was taken into account, adjusting the model to these dimensions.

Density contrast: gravity inversion is not sensitive to any layers stratification, so density contrast are only relative to the stratum in which it is contained. For the whole subsurface volume, we selected the possible density contrasts by means of the extreme values, −250 kg/m^3 and +250 kg/m^3. These values were selected empirically, looking for connected and well-developed subsoil volumes. Previous studies used similar density contrast values (e.g. Araña et al., 2000). Nevertheless, the choice of these values is not critical; indeed, even if different density contrasts are considered, the resulting solutions are similar. However, for higher contrasts, smaller volumes would be obtained.

Regularization parameter β: it is widely used in geophysical inversion methods (e.g., Zhdanov, 2002; Farquharson and Oldenburg, 2004). Lower β values, provide best fitted solutions but increase the anomalous volumes. High β values lead to smaller or shallower models. Aiming to choose a consistent value for β, the solution adopted searches for a value which supplies that the first term of the error function is similar to the standard deviations of the data dispersion or, alternatively, which provides the best fit to the data under the imposed constraints. The best solution in this case corresponds to a value of 0.3.

Initial population: initially, the process generates a population of distinct models (individuals), randomly changing the density contrast

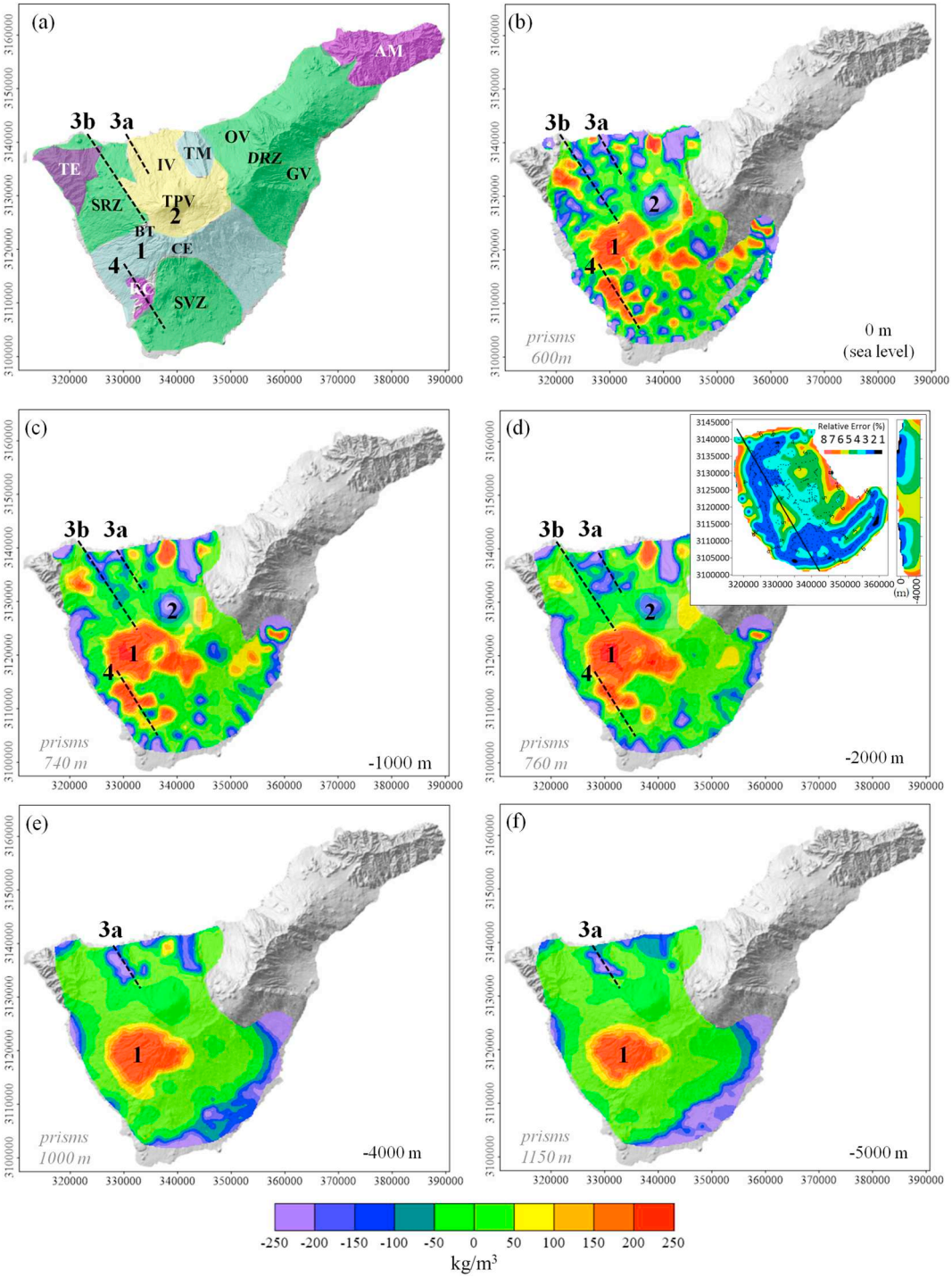

Fig. 4. (a) Main geological features of Tenerife. They are highlighted using the same abbreviations as in Fig. 2. (b), (c), (d), (e), (f). Different horizontal slices obtained from the 3D density model developed using a genetic inversion procedure from 0 m to 5000 m b.s.l. The cell sizes of the 3D density model are indicated at the bottom left corner of the corresponding image at depth. The main anomalies are numbered from 1 to 4. (1) Central main height density body, (2) low density contrast under Teide-Pico Viejo Complex, (3) low density contrast under Santiago Rift, following the NNE-SSW orientation, and (4) a branch from the central structure under the Roque del Conde area. Panel (d) (inset) shows the relative error in relation to the root mean squared value for the residuals (see text for details).

of some cells. The size of the population, np, which remains constant throughout the process, is empirically chosen taking into account parameters such as number of prisms or data set. This parameter has influence on the convergence speed, but it does not affect to final model obtained (Michalewicz, 1994). Taking into account the average size of the variables acting in this process, we used an initial population with ten individuals, which let us obtain the evolution of a significant population through a fast process.

At every stage of the iterative process, individuals of the corresponding population were evaluated and chosen using a selection operator (Michalewicz, 1994). The GA selects the individuals that best satisfy the minimization condition. Then, two genetic operators (mutation and crossover) generated new individuals, which were evaluated. The mutation operator provides random diversity by randomly modifying the density contrast of some prisms of some models. Furthermore, the crossover operator acts on and exchanges information

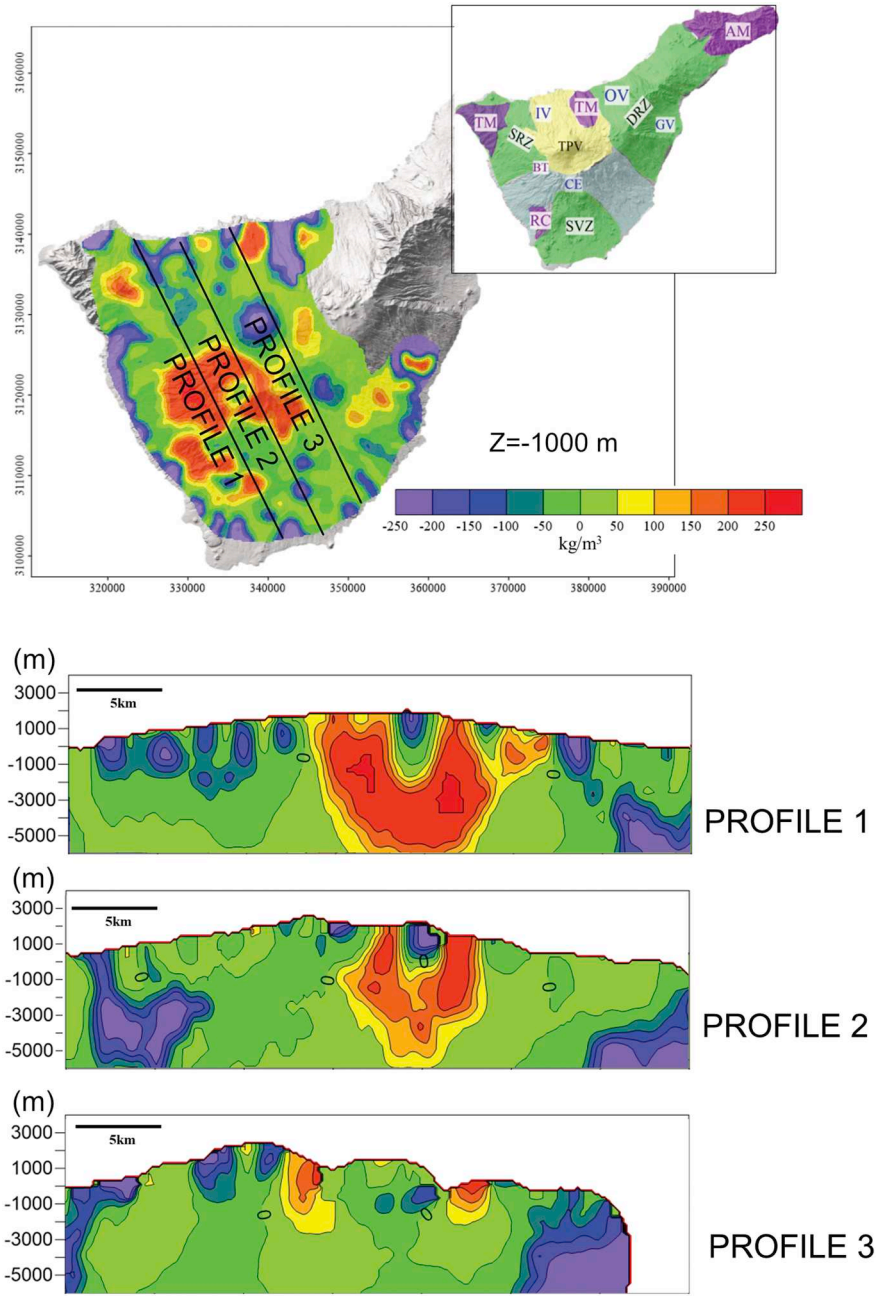

Fig. 5. Comparison between different vertical cross-sections showing the density contrast structure at depth along three different profiles (see text for more explanation) The main geological features are included as an inset (see also Fig. 1) (see text for discussion).

between several selected models. Then, the modified models were evaluated again with respect to the unmodified ones, then repeating the evolutionary process with the best models.

This iteration process continued until a predetermined value for the error function was reached. After several trials, taking into account the disperse distribution of the data associated with the very rugged topography, we preferred to establish a reasonably conservative value of 400 μGal as limit value to each stage, despite the estimated value of the anomalies gravity data set was around 200 μGal.

Once this value was attained, a new stage of the process began and a model with smoother geometry was searched (Montesinos et al., 2005). In order to remove isolated prism with non-null density contrast or small holes within the source a smooth operator was implemented. It removed these small discontinuities in the volume and built a smoother geometry. When the fit of the best individual which was smoothed reached values next to the limit value or it did not change in successive stages, the process ended. In this case, the iterative procedure was repeated 7 times with a total of 707,040 generations, reaching a fitting value of the gravity anomalies of 0.553 mGal (rmse). At this point, the process finished and the best 3D density contrast model of the subsoil was taken as the solution of the gravity inverse problem.

After applying the inversion algorithm explained above, we obtained a density value for each of the prisms of the model. In our case we obtained values for prisms located between 500 m a.s.l. down to 13,500 m b.s.l. The smaller size prisms (approximately $600 \times 600 \times 600$ m^3) coincide with those found in the upper part of the model, at about 500 m a.s.l. On the other hand, the dimensions of the prisms of the lower parts of the model shown in this work (approximately 6000 m b.s.l.) are around $1200 \times 1200 \times 1200$ m^3. We did not include results below 6000 m b.s.l. since it is not the objective of

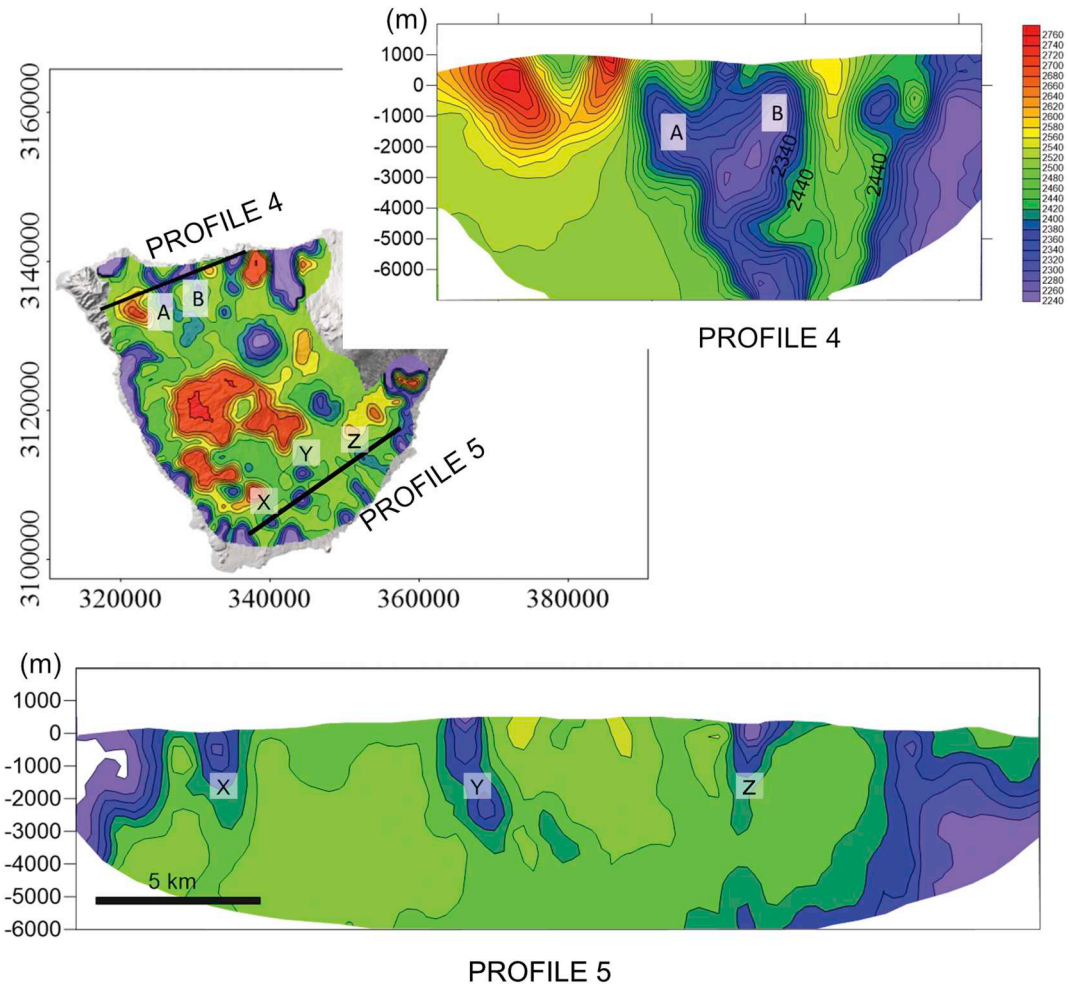

Fig. 6. Additional profiles at the north (Profile 4) and south (Profile 5) of the island showing the density contrast structure at depth (see text for discussion).

this study and the results are similar to those of previous works. Different views of the final model are shown in Figs. 3 to 6.

In order to understand how the GA resolves the structures at depth, we evaluated both the sensitivity of the data and the model:

On the one hand we evaluated the sensitivity of the inversion to our data set. It was determined by the MxM matrix $\mathbf{C_M}$ where the each jj-element depends on the position of each j-cell and the distribution of the stations and the error of the data. In this case, we determined the relative error in the modelling of the different structures at that depth in relation to the root mean square value for the residuals of the inversion process. So, this error is calculated from the diagonal elements of $\mathbf{A}^{\mathbf{T}}\boldsymbol{E}_{ss}{}^{-1}\mathbf{A}$, divided by the r.m.s.e obtained to fit the gravity data. As it would be expected, the main influence in the observed gravity values mainly depends on the cells under the vertical of the stations. We obtain values around 2 to 10% (high sensitivity) regarding to the total error of the fitting until 5 km b.s.l. being greater in the zones with a minor coverage of points at surface (low sensitivity). Inset in Fig. 4d shows this relative error calculated at 2000 m.b.s.l as it is mentioned above.

On the other hand, to evaluate the sensitivity of the model, we studied the solution obtained using the GA for a synthetic example with a similar geometry or location to the sources and the same field data distribution. Thus, we were able to evaluate how the GA resolves areas covering by a less amount of data, such as the central structure of high density contrast, the branches that are observed from 2000 m b.s.l. This example also allows testing the GA resolution capacity for the alignments of low density structures observed in the northwest rift (Fig. 7). In the same way, it could be shown how GA may reproduce other more specific synthetic models, as it can be seen in previous works (e.g. Montesinos, 2002). Here, we have preferred to show how a model with similar characteristics to the final density model obtained, where small changes in the geometry of the model are introduced, can be reproduced. We consider that the inputs imposed in this simulated model (mainly density contrasts) are enough to assure the reliability of the final model.

For additional information concerning the methodology used, we refer the reader to previous papers (e.g., Montesinos, 2002; Montesinos et al., 2005, 2006) that specifically describe it, with details on its sensitivity and accuracy, including some examples of its applications.

6. Results

The local anomaly map (Fig. 2) obtained by this study after removing a regional component, shows features similar to those reported in previous studies (MacFarlane and Ridley, 1968; Ablay and Kearey, 2000; Araña et al., 2000; Gottsmann et al., 2008). The most striking feature is the large, high gravity anomaly, almost circular in shape, centered beneath the southwest of the caldera of Las Cañadas (Fig. 2), which dominates a large part of the island of Tenerife. Previous studies explained this anomaly as due to a large buried basaltic volcano known as Boca Tauce (Ablay and Kearey, 2000) that forms part of the basaltic shield edifice.

In terms of the 3D gravity inversion, the increase in amount of field data, the narrower density contrasts range, and the precise terrain correction used in our calculations, we were able to improve the resolution of certain internal structures, particularly at shallow depths under the Santiago del Teide rift, the central part of the island, and in

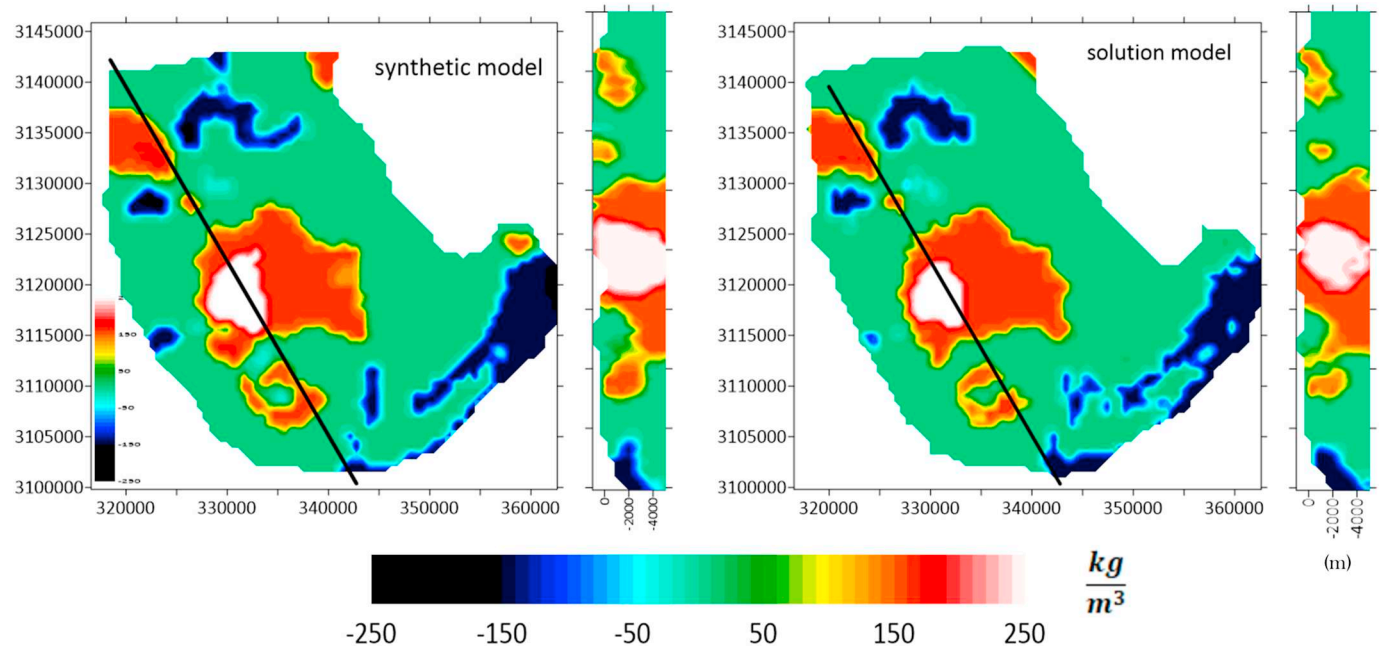

Fig. 7. Example of the sensitivity analysis to uncertainties in the sources position and their geometries. We obtained a synthetic model composed by a central massive body with a branch that reaches the surface, and several low density bodies to the NW of the model. The forward problem provides the gravity field generated by this synthetic example in the same gravity stations where the observations located on Tenerife island were made. We show the result in the section to 1000 m b.s.l and with a vertical section along the profile indicated in the figure. As it can be observed, most of the main features of the synthetic model are properly reproduced by the genetic algorithm.

southern Tenerife. Concerning the central part of Tenerife, the main high-density body extends from the maximum depths reached by our model (around 13 km b.s.l.) to shallower levels at around 500 m a.s.l. (numbered Area 1 in Fig. 4). This is the main structure that produces the observed high residual anomaly in the southwestern part of Las Cañadas caldera (Fig. 2). This body extends towards the surface via branched paths that extend upwards from a depth of around 2 km b.s.l. The areas of Roque del Conde and Boca de Tauce are the only visible signs on the surface of this structure (Ablay and Kearey, 2000; Gottsmann et al., 2008) (see Fig. 1a). Additionally, the model reveals the presence of a significant low-density contrast under the Teide-Pico Viejo stratovolcanoes (numbered Area 2 in Fig. 4), which has minimal values of around 2–3 km b.s.l., as shown in Figs. 4 and 5.

Aside from the areas occupied by the main high-density body and low-density contrast under the Teide-Pico Viejo stratovolcanoes, our model also shows other remarkable density contrasts in the sectors occupied by basaltic volcanism. To better visualize the main density contrasts of our 3D density model related to the Santiago del Teide rift and the southern volcanic area on Tenerife, we plotted them in several horizontal (Fig. 4) and vertical (Figs. 5 and 6) sections at different depths and following a parallel arrangement, respectively. This allowed comparing these two areas that host recent basaltic volcanism and to establish similarities and differences in their internal structures. From Fig. 4 we observe a variation in the depth of the low-density anomalies representing the zones with basaltic volcanism. A significant low-density anomaly (numbered 3a in Fig. 4) is clearly discernible below the western border of the Icod Valley, which is persistent from 0 m to a depth of > 5000 m. This anomaly, identified by Gottsmann et al. (2008) as a main intrusion zone for basaltic magmas, was related to the periodic ground deformation detected by DGPS and InSAR observations (Fernández et al., 2005) and to the unrest episode that took place on the island in 2004 (Pérez et al., 2005; García et al., 2006; Gottsmann et al., 2006; Almendros et al., 2007; Martí et al., 2009; Dominguez-Cerdeña et al., 2011). Using our data, it is now possible to distinguish the shape of this structure in greater detail: it is rooted 5 km b.s.l. and is aligned in a NNW-SSE direction (Profile 2 in Fig. 5), thus forming a narrow angle in relation to the alignment of the cones in the Santiago del Teide rift observed at the surface (Line 3b in Fig. 4; Profile 1 in Fig. 5). This anomaly is maintained at shallower depths while changing a little bit in its extent (see Fig. 4(b, c, d)), but still keeping its orientation parallel to the trajectory of the main regional compressive stress in Tenerife, according to Geyer et al. (2016). In the zone corresponding (at the surface) to the Santiago del Teide rift (Line 3b in Fig. 4) we observe an alignment of continuous low-density contrasts rooted at around 2 km b.s.l. These density contrasts seem to join with those from the western sector of the Icod valley at a depth of 1–2 km b.s.l. (see Fig. 4c, d, e). This connection between the two groups of gravity anomalies is also visible in the vertical profiles (e.g., Profile 4 in Fig. 6).

Below the southern volcanic zone, similar shallow density contrasts appear in the areas occupied at surface by the monogenetic volcanoes (Fig. 6, Profile 5). Furthermore, the continuation of the dense core of the basaltic shield is visible below the Roques de García area (Profile 4 in Fig. 4) anomalies. Isolated low-density contrasts, similar to those observed below the Santiago del Teide rift, appear at shallow depths (1–2 km, Fig. 4). However, they do not define a clear lineation such as what occurs below the rift. A much larger and deeper (> 5 km) anomaly appears at the southern side of Tenerife, directly reaching the surface at some points (Profile 5 in Fig. 6), and approaching the shallower anomalies in other zones (Profile 2 in Fig. 4).

7. Discussion and conclusions

Our results confirm previous gravimetric studies and improve the understanding of the shallowest part of the Tenerife basaltic plumbing system. On the one hand we confirm the observations made by Gottsmann et al. (2008) that the gravity anomaly, which is clearly discernible below the western border of the Icod Valley, may correspond to a main intrusion zone for basaltic magmas. On the other hand, they reveal some differences in their internal structures below the northwestern and southern sectors of the island. In addition, they shed light on how rift systems distribute magma within the shallowest part of the island, and confirm observations from other large oceanic volcanic

islands such as Hawaii (e.g., Polland et al., 2014) and Reunion (e.g., Michon et al., 2015), in the sense that rift zones are shallow structures that are used by deep basaltic magmas to distribute themselves inside the volcanic edifice when they reach shallower depths. Our results also provide new details on the internal structure of the central part of Tenerife.

Our data suggest that the Santiago del Teide rift only represents the shallowest part of the basaltic plumbing system in that sector of the island, where a set of aligned low-density contrast bodies appear below the lineation of volcanic cones observed at the surface. Below the rift zone these density anomalies extend from the surface to a depth of no > 2–3 km b.s.l, but seem to connect throughout most of its extension with a much deeper and larger anomaly located at the western border of the Icod valley that follows a NNW-SSE direction (Fig. 4, d); this coincides with the trajectory of the main regional compressive stress on Tenerife (Geyer et al., 2016) and also with one of the two main alignments identified in southern Tenerife (Buchner and Kröchert, 2009). This anomaly has already been identified by Gottsmann et al. (2008) as representing a main pathway for the ascent of deep basaltic magmas. Also, this trend coincides with a clearly marked P-wave anomaly trend found in the same area using seismic tomography (García-Yeguas et al., 2012). This NNW-SSE direction coincides with one of the main structural trends identified in the Canaries that could be related to regional structures already present in the basement (Geyer and Martí, 2010; Geyer et al., 2016). The same orientation is observed above the high-density anomaly in the Roque del Conde area, which suggests that this regional trend also controlled the preferential intrusive paths in earlier stages of the construction of the island. According to this, we suggest that this regional structure, located at the western side of the Icod valley, corresponds to one of the main ascent pathways of basaltic magmas in the north of Tenerife. When magmas reach a depth of 2–3 km, they are captured by the rift structure and transported laterally inside the volcanic edifice, propagating in multiple directions (e.g., Soriano et al., 2008; Delcamp et al., 2014) and reaching the surface through the weakest points of the rift. The similarities between the Santiago del Teide and the Dorsal rifts (e.g., the constant alignment defined by the location of the volcanic vents and a nearly constant orientation of dykes parallel to the alignment of the cones), as well as the identification of a principal lateral flow component in the dykes (Delcamp et al., 2012, 2014), both suggest that the Dorsal rift has a similar shallow character. There is also good agreement with recent magnetotelluric results that have described the shallow nature of these rift structures (Pous et al., 2002; Piña-Varas et al., 2014). However, from the results obtained in this study we cannot identify which channel would be the deep feeding channel of the Dorsal rift. The structure of this plumbing system coincides with the storage of basaltic magma envisaged by Albert et al. (2015) using a petrological analysis of historical eruptions on Tenerife. In addition, the predicted lateral shallow transport of magma through the rift zones is corroborated by existing AMS data that reveal a dominant lateral flow pattern in the Dorsal rift (Delcamp et al., 2014).

In contrast, the southern volcanic zone on Tenerife does not show the same structural pattern at surface as the rift zones. Instead of showing a single preferential alignment of cones, it is characterized by scattered vents or clusters of vents with no constant alignment of their eruption fissures; nevertheless, three main alignments can be discerned, namely NW-SE, NNW-SSE and NE-SW (Buchner and Kröchert, 2009; Geyer and Martí, 2010). In a similar way, the distribution of the anomalous density bodies resulting from our inversion model does not show a unique persistent lineation in southern Tenerife (Fig. 4b). Conversely, they correspond to shallow isolated bodies that in some cases connect to much deeper, similar density anomalies and that reflect the position of the main age clusters of eruptive vents identified by Kröchert and Buchner (2008). Therefore, we can conclude that the shallow structure of the basaltic plumbing system differs between the Santiago del Teide rifts and the southern volcanic zone, probably as a result of the existence of different tectonic controls in each zone—-particularly in the shallowest part of the volcanic system. This could be due to different gravitational stresses and structural configurations of the pre-volcanic basement, but also to its own distribution of shallow stresses associated with the two rift systems, as suggested by Geyer and Martí (2010). Similar distributions have been described at Reunion (Michon et al., 2015) and Hawaii (Lockwood and Lipman, 1987). However, the situation in terms of deep basaltic magma feeding in the NW rift and the southern volcanic zone is similar, as in both zones basaltic magma seems to come from peripheral paths, being then captured by shallow fractures that control its final ascent to the surface. What we interpret for each zone is that this shallow fractures system is different, as was already pointed out by Geyer and Martí (2010).

In our model, the resulting density anomalies underneath the Santiago del Teide rift and the southern monogenetic volcanic field, which we interpret as pathways for basaltic magma to reach the surface, are negative. In contrast with other studies where such kinds of anomalies appear positive (e.g., Kauahikaua et al., 2000; Park et al., 2007). However, negative density contrasts have also been detected under other basaltic rift zones on ocean volcanic islands, such as the NE rift zone of Piton de la Fournaise volcano (Gailler et al., 2009) or in other islands from the Canarian Archipelago (Camacho et al., 2019). The mean density and the extreme density contrasts used are the critical points in order to define positive and negative density contrasts. Thereunder, the negative contrast under the Santiago del Teide rift means that such zones are less dense than the surrounding old basaltic shield. We interpret these basaltic pathways as highly fractured zones infilled with highly fractured basaltic rocks, as can be observed for some shallow dykes that crop out at the rifts zones (Soriano et al., 2008), and therefore of lower density than the intruded host rock.

In addition to the observations made on the structure of the shallowest part of the basaltic plumbing system, our results also provide new light on other relevant aspects of the internal structure of Tenerife. The presence of a high-density body in the center of the island confirms a number of existing interpretations (Ablay and Kearey, 2000; Araña et al., 2000; Gottsmann et al., 2008), who already postulated the continuation of the basaltic shield structure below the current central volcanic complex formed by the Cañadas caldera and the Teide-Pico Viejo stratovolcanoes. Such a configuration supports the accumulation of dense magma at the interior of the central part of the island, but not the hypothesis that the evolution of Tenerife involved the construction of three isolated proto-islands (Teno, Anaga and Roques del Conde), the space in between which was subsequently occupied by the central complex (Ancochea et al., 1990; Carracedo, 1994; Walter, 2003).

The low-density contrast located below the Teide-Pico Viejo and Las Cañadas caldera (numbered 2 in Fig. 4) is interpreted to be the combined effect of i) low-density phonolitic materials filling the collapsed sector of the Las Cañadas edifice (Martí et al., 1994), ii) the infill of the caldera depression by poorly consolidated, highly porous and permeable lavas from Teide and Pico Viejo (Martí et al., 2008), iii) the presence of a deep aquifer inside the caldera depression (Coppo et al., 2008), iv) the existence of an active hydrothermal system (Piña-Varas et al., 2014; Villasante-Marcos et al., 2014), and v) the presence of fresh magma accumulated at shallow depths (Ablay et al., 1998; Andujar, 2007). This latter contributing factor is consistent with the results obtained by the array beam-forming method that reveals possible fractures filled by volatiles or magma at 2–4 km b.s.l. in the Las Cañadas caldera (Barros et al., 2012). However, with the available data it is not possible to discern which of these contributing effects was the most important in defining the observed gravity low. In the same sense, other available geophysical data (Pous et al., 2002; Prudencio et al., 2013, 2015; Piña-Varas et al., 2014, 2015) do not allow discriminating between the particular effects of each of these contributors due to the low-density contrast. However, it is worth noting that all the available geophysical data, as well as the new gravimetric data, agree on the limits of the caldera depression that extend below the Teide-Pico Viejo

stratovolcanoes.

In summary, our results confirm certain relevant aspects of the internal structure of Tenerife—in particular relative to the shallow part of the basaltic plumbing system. The rift systems only control the movement and distribution of basaltic magmas when they reach very shallow levels inside the volcanic edifice and only in the sectors affected by these extensional structures, which may result from a combination of gravitational loading and unloading, volcano spreading, and regional and local tectonics. On the other hand, the presence of an extensive monogenetic volcanic field at the southern part of the island responds to a different shallow structural control, not being affected by the shallow rift systems. These results indicate that the coexistence of shallow rift system monogenetic fields showing a much wider distribution of volcanic centers is compatible with large ocean island basaltic volcanoes. This implies that magma ascent and eruption may have different structural controls depending on the sector of the volcanic edifice we are considering, so precise knowledge of regional and local tectonics and of the internal distribution of rock bodies is crucial from the point of view of eruption forecasting in such volcanic environments.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.pepi.2019.106319.

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